

Exercise Sheet 1

Advanced Microeconomics: Economic Decision Making

Question 1

For a rise in p_1 from p_1^0 to p_1^1 (other prices, wealth fixed), define the positive loss measures $CV = e(p^1, u^0) - w$, $EV = w - e(p^0, u^1)$, $\Delta CS = \int_{p_1^0}^{p_1^1} x_1(p_1, w) dp_1$, with $u^0 = v(p^0, w) > u^1 = v(p^1, w)$. Suppose good 1 is normal in the sense that $h_1(p, u)$ is increasing in u .

- (a) Express CV and EV as integrals of Hicksian demand. [8]
- (b) Prove $EV \leq \Delta CS \leq CV$. [12]
- (c) State what the sandwich collapses to when wealth effects on good 1 vanish, and why the width of the sandwich is informative. [5]

Question 2

A consumer's EMP is $\min p_1 x_1 + p_2 x_2$ s.t. $x_1^{-1} + x_2^{-1} \leq u^{-1}$.

- (a) Solve the EMP: derive $h(p, u)$ and show $e(p, u) = u(\sqrt{p_1} + \sqrt{p_2})^2$. [9]
- (b) Let $w = 12$, $p^0 = (1, 1)$ and let p_1 rise to 9. Compute u^0 , u^1 , CV , EV and ΔCS , confirming the sandwich ordering. (*Roy's identity gives $x_1(p, w) = w/[\sqrt{p_1}(\sqrt{p_1} + \sqrt{p_2})]$.*) [10]
- (c) Compute the own-basket payment and its excess over CV as a share of the payment; compare with Question C1's share and explain the difference in one sentence. [6]

Question 3

Call $\succsim$ homothetic if $x \sim y$ implies $\lambda x \sim \lambda y$ for all $\lambda > 0$, and normalise $u(\lambda x) = \lambda u(x)$.

- (a) Show that z solves the EMP at target 1 if and only if uz solves the EMP at target $u > 0$; deduce $e(p, u) = u e(p, 1)$ and $h(p, u) = u h(p, 1)$. [12]
- (b) Deduce that all income elasticities of demand equal one, and that wealth-expansion paths are rays. [8]
- (c) A price index for a homothetic consumer is proposed: $P = e(p^1, 1)/e(p^0, 1)$. Show that compensation by the factor P is *exact* for this consumer: $e(p^1, u^0) = P \cdot w$. [5]